

A Model details

Given the grid-level histology feature h_i , SpaceRec predicts two outputs for each retained grid i : grid-level gene expression $\hat{\mathbf{y}}_i$ and grid-level cell-type proportions $\hat{\mathbf{p}}_i$. The histology feature is first mapped to a shared latent representation,

$$z_i = f_\psi(h_i),$$

where f_ψ is a multilayer perceptron. The **expression head** predicts a nonnegative gene-expression vector,

$$\hat{\mathbf{y}}_i = \text{Softplus}(f_{\text{expr}}(z_i)), \quad \hat{\mathbf{y}}_i \in \mathbb{R}_{\geq 0}^M,$$

where M is the number of modeled genes. The **cell-type head** predicts a grid-level proportion vector,

$$\hat{\mathbf{p}}_i = \text{Softmax}\left(\frac{f_{\text{ct}}(z_i)}{T}\right), \quad \hat{\mathbf{p}}_i \in \Delta^{C-1},$$

where C is the number of modeled cell types and T is the softmax temperature.

Let $\mathcal{B}_s$ denote the set of grids assigned to Visium spot s . SpaceRec links grid-level predictions to spot-level supervision by aggregating the grids inside each spot. Because expression is additive, the reconstructed spot-level expression is computed as

$$\hat{\mathbf{y}}_s = \sum_{i \in \mathcal{B}_s} \hat{\mathbf{y}}_i.$$

In contrast, cell-type proportions are averaged within the spot:

$$\hat{\mathbf{p}}_s = \frac{1}{|\mathcal{B}_s|} \sum_{i \in \mathcal{B}_s} \hat{\mathbf{p}}_i.$$

Thus, $\hat{\mathbf{y}}_s$ is compared with the observed raw Visium expression $\mathbf{y}_s$, while $\hat{\mathbf{p}}_s$ is compared with the RCTD-derived spot-level proportion vector $\mathbf{p}_s$.

The expression reconstruction loss is defined using a Huber loss after applying a $\log(1+x)$ transformation to both reconstructed and observed raw expression:

$$\mathcal{L}_{\text{expr}} = \frac{1}{|\mathcal{S}|M} \sum_{s \in \mathcal{S}} \sum_{m=1}^M \ell_\delta(\log(1 + \hat{y}_{s,m}), \log(1 + y_{s,m})),$$

where $\mathcal{S}$ is the set of training spots and ℓ_δ is the Huber loss with threshold δ . The log transformation is used only inside the loss; grid-level expression predictions are still summed on the original additive scale.

The deconvolution loss matches the aggregated grid-level cell-type prediction to the RCTD-derived spot-level target:

$$\mathcal{L}_{\text{deconv}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} D_{\text{KL}}(\mathbf{p}_s \| \hat{\mathbf{p}}_s).$$

Because spot-level composition supervision alone may allow diffuse grid-level predictions, SpaceRec includes a confidence regularization term that encourages each grid to assign higher probability to a dominant cell type:

$$\mathcal{L}_{\text{conf}} = -\frac{1}{G_{\text{spot}}} \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{B}_s} \log \left[\max \left(\max_c \hat{p}_{i,c}, \epsilon \right) \right],$$

where $G_{\text{spot}} = \sum_{s \in \mathcal{S}} |\mathcal{B}_s|$ is the total number of grid observations included in the spot-level training bags, and ϵ is a small constant for numerical stability.

The final training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{expr}} + \lambda_{\text{ct}} [(1 - \alpha)\mathcal{L}_{\text{deconv}} + \alpha\mathcal{L}_{\text{conf}}],$$

where λ_{ct} controls the overall contribution of cell-type supervision relative to expression reconstruction, and $\alpha \in [0, 1]$ controls the balance between matching RCTD-derived spot proportions and encouraging concentrated grid-level predictions.